

CMT Project

Migration Modeling for South America

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1 Deviation from project proposal

Initially, the project aimed to develop a model capable of predicting emigration flows (outward migration) from a country, based on a limited set of high-impact factors. The variables originally envisaged were GDP, unemployment, political stability, and climate exposure, while leaving open the possibility of including diaspora size (migrant stock) or climate shocks. The initial idea was to consider data from 2000 to 2019 in order to predict future years, using global data and thus remaining applicable to any country worldwide, with the expectation of being able to compare results with empirical data and existing advanced models.

During the development of the project, the first major structural change was the decision to work with Net Migration data (the difference between immigration and emigration), due to the substantially easier access to data for this variable compared to the difficulty of obtaining reliable data on emigration alone. Subsequently, it became clear that different parts of the world are characterized by distinct macro-level social dynamics, which led to the introduction of a regional focus: South America, comprising 12 countries. In order to ensure both satisfactory data availability and strong regional relevance, the following variables were selected: GDP, unemployment, average years of schooling (for the population aged 25 and over), and homicides per 100k habitants.

The temporal focus and overall structure of the model also changed. The model now links empirical net migration data from 1990 to 2014 in order to estimate the relationships between net migration and the selected factors, and then uses these relationships to predict net migration values for the period 2015–2019. The forecasting period was bounded to 2019 to exclude the immense impact of the COVID-19 pandemic on migration.

With regard to the evaluation of results, the original idea of comparing outcomes with advanced pre-existing models was abandoned, given the fundamental differences in model structure and the deliberately limited number of parameters considered. Instead, the predicted results are compared exclusively with empirical data.

2 Introduction to the problem

International migration is one of the most persistent and consequential socio-economic phenomena in the contemporary world, particularly in developing and emerging regions, where migration flows often respond to rapid and asymmetric economic, social, and institutional changes. In South America, these dynamics are especially complex due to strong cross-country heterogeneity in economic growth, labor market conditions, educational attainment, and public safety. Such heterogeneity complicates both the interpretation of observed migration patterns and the construction of reliable forecasting tools.

From an empirical perspective, migration analysis is further challenged by the limited availability of homogeneous and long-run data, a constraint that is particularly binding for many middle and low-income countries. In these settings, highly complex or data-intensive models are often impractical, difficult to interpret, and poorly transferable across contexts. This project therefore deliberately adopts an alternative approach, relying on a small set of key variables relatively “simple” yet theoretically and empirically relevant that capture the main economic and social dimensions underlying migration decisions.

Choosing a simple model serves two main purposes. First, it makes the results transparent and easy to explain. By not over-complicating the math, we can clearly see the link between our

findings and real-world migration theories. Second, it makes the model practical and flexible. Because it does not require a lot of complex information, it can still produce reliable results in countries where data is hard to find or analytical tools are limited. This approach ensures that our model is both easy to understand and useful for a wide range of different regions. In this sense, the project aims not only to explain migration patterns in South America, but also to propose an essential and replicable empirical framework that can support evidence-based analysis and the design of informed migration policies.

3 Approach

3.1 Data collection and preprocessing

The empirical analysis relies on annual country-level data for South America. Net migration is treated as the dependent variable, observed over 1990–2014 for model estimation and predicted over the period 2015–2019. The explanatory variables include GDP per capita, unemployment rate, average years of schooling, and homicide rates, capturing key economic, social, and security-related conditions relevant to migration dynamics. Net migration [17], GDP [18], unemployment [19], and homicide [16] data were obtained from the World Bank World Development Indicators, while schooling data [20] were sourced from the United Nations Development Programme (UNDP). All datasets cover the twelve South American countries over the period 1990–2019.

The choice of variables follows standard migration theory and reflects region-specific dynamics. Economic conditions and labor market performance are captured through GDP and unemployment, long-term outlook and future possibilities through years of schooling, and public safety through homicide rates. In the South American context, homicide rates provide a more relevant proxy for insecurity than conflict-based indicators, as large-scale armed conflicts are rare, while criminal violence constitutes a persistent driver of migration [5].

Raw World Bank CSV files present several practical issues, including irregular or multi-line headers, empty rows, misaligned columns, byte-order marks, and heterogeneous country naming conventions. These formatting inconsistencies vary across indicators and make direct data import unreliable. To address this, each dataset was manually parsed and cleaned, with explicit control over header reconstruction, column alignment, country filtering, and year selection. As a result, the preprocessing scripts differ in length and structure across variables, reflecting the heterogeneous formatting of the raw files rather than differences in conceptual treatment.

All variables were reshaped into a common panel format with years as rows and countries as columns. This choice ensures consistency across datasets and simplifies their integration into the modeling pipeline. Using a unified structure makes it straightforward to reshape the data into long format, merge variables on Country–Year pairs, and apply the same regression and feature engineering procedures across countries.

The UNDP schooling data required less extensive preprocessing, as they were already provided in a structured country–year format. However, they were still reshaped to match the common panel structure used for the other indicators. This ensures that schooling enters the interpolation, feature engineering, and regression steps in exactly the same way as the remaining variables.

3.2 Prediction model and forecasting strategy

First, missing values were processed using variable-specific interpolation: linear interpolation for GDP, piecewise cubic hermite interpolating polynomial (PCHIP) for schooling to ensure a

smooth curve, and cubic splines for unemployment and homicide rates to capture cyclical fluctuations.

The modeling strategy employs feature engineering to capture the non-linear and dynamic nature of migration. Explanatory variables were transformed based on specific theoretical justifications. Logarithmic transformations were applied to GDP, schooling, and homicide rates to account for diminishing marginal returns; for instance, a unit increase in income has a larger impact on migration decisions at lower income levels than at higher ones [6]. Derivatives were calculated to capture trends: GDP Acceleration (the second derivative) detects economic shocks, while Unemployment Change highlights deteriorating labor conditions rather than static levels. Interaction terms were constructed to model compound risks, such as the Safety Index (Homicide x Unemployment), reflecting the hypothesis that physical insecurity drives migration more intensely when coupled with a lack of economic opportunity [5].

To estimate the migration function, the analysis employs MATLAB’s Stepwise Linear Regression `stepwiselm` [1] using the Bayesian Information Criterion (BIC) for variable selection. Given the limited sample size (25 annual observations per country) relative to the large pool of engineered features (15 potential predictors including lags and interactions, explained in Appendix-A), a standard Ordinary Least Squares (OLS) regression including all variables led to severe overfitting. The model captured random noise rather than structural trends, resulting in poor performance for the test period.

The stepwise algorithm addresses this by iteratively constructing a model for each country. It begins with a "null" model (containing only an intercept) and successively tests potential predictors from the defined "upper bound" formula. At each step, the algorithm adds the most statistically significant variable or removes a variable that has become redundant, trying to minimize the BIC score. BIC imposes a penalty on model complexity. This ensures parsimony, favoring a simple model with fewer, highly impactful variables over complex models that might fit the training data perfectly but fail to generalize.

In order to assess the performance of the model on the training data (1990–2014), two different R^2 values were computed. The R^2 indicates the share of net migration variance explained by the model, but it may be misleading since it increases mechanically with the number of explanatory variables. A high R^2 may therefore reflect an overly complex model rather than genuine predictive efficiency. The **adjusted** R^2 corrects for this effect by increasing only if the new variable improves the model enough to compensate its cost, which penalizes model complexity. Its mathematical formula is

$$R_{adj}^2 = 1 - \frac{\text{MSE}}{\text{Var}(y)}$$

A small gap between R^2 and adjusted R^2 suggests that the selected variables add genuine explanatory power without excessive overfitting. Finally, the RMSE (Root Mean Squared Error) measures the average prediction error in absolute terms, expressed as the number of migrants.

3.3 Results control

The objective of the results control phase was to rigorously evaluate the migration model’s performance by comparing the predictions of our model (`predicted_migration_all_countries.csv`) against the actual measured data (`NetMigration_SouthAmerica.1990_2019.csv`). This validation process was designed to identify not only the magnitude of prediction errors but also the nature of these deviations - specifically, whether the model exhibits systematic bias or struggles

with specific temporal trends. The control workflow consists of two distinct layers: quantitative statistical analysis (implemented in C) and visual diagnostic reporting (implemented in MATLAB).

3.3.1 Selection of Error Metrics

To capture the nuances of migration data, which often varies by orders of magnitude between countries, a multi-metric approach was adopted:

- **Mean Absolute Error (MAE):** Used as the primary accuracy indicator to measure the average error magnitude. It provides a direct assessment of how far predictions typically deviate from reality.
- **Mean Bias Error (MBE):** By preserving the error sign ($Actual - Predicted$), this metric identifies systematic bias, indicating whether the model consistently over- or under-predicts migration in specific regions.
- **Median Absolute Error:** Selected as a robust alternative to the mean, this measure reduces the influence of extreme outliers or singular “shock” years on the overall assessment.
- **Mean Absolute Percentage Error (MAPE):** This standardizes error relative to total net migration. It is essential for comparing performance across countries with vastly different migration volumes, ensuring large nations do not overshadow smaller ones.

3.3.2 Visualization Strategy

A comprehensive plotting strategy was employed to visualize the results:

1. **Global Summary Analysis (Logarithmic Scale):** A summary dashboard was generated to compare error metrics across all countries simultaneously. Given the vast disparity in migration numbers, a *logarithmic scale* was necessary for the MAE and MBE charts. This allows for the simultaneous visualization of errors in major destination countries alongside smaller nations without the latter being rendered invisible. Percentage Error values exceeding the 190% graph limit are truncated, potentially underrepresenting the true error for outliers.
2. **Temporal Trend Analysis (Split-Layout):** For granular validation, individual diagrams were generated for each country.
 - *Real vs. Predicted:* This line graph visualizes the trajectory of the model over time, helping to identify if the model captures turning points (e.g., the onset of a migration surge) or simply follows a linear trend.
 - *Error Evolution:* A bar chart displaying the percentage error per year was placed directly below the trajectory. This isolates the error component, making it visible if relative accuracy is improving or degrading over time.

This methodology ensures a holistic validation: the C code handles the computational heavy lifting of processing raw inputs into robust statistics, while the MATLAB visualization layer translates these statistics into actionable insights regarding model behavior.

4 Results

4.1 Model Fit and Explanatory Power

The R^2 and **adjusted** R^2 values computed on the training period (1990–2014) indicate that the model explains a substantial share of the variation in net migration for most countries. Indeed, most R^2 values are above 0.90 while the lowest is 0.70. This suggests that the selected economic, social, and security variables capture key structural drivers of migration unexpectedly well.

However, a high in-sample R^2 does not necessarily imply accurate out-of-sample predictions. In some cases, strong training performance coexists with larger prediction errors after 2014, particularly when migration dynamics are affected by unexpected shocks such as economic crises, conflicts, or major political changes that are not explicitly modeled.

The generally small gap between R^2 and **adjusted** R^2 indicates that model performance is not driven by excessive complexity, but rather by variables that contribute meaningfully to explanatory power, which was the main purpose of this project. It succeeded in using a small set of parameters that have an important influence on the model, as the highest difference between R^2 and **adjusted** R^2 is 0.045. Overall, the results confirm good in-sample fit while highlighting the inherent limitations of forecasting migration in the presence of structural breaks.

4.2 Model performance

The result controls show that the predictions are more accurate at the beginning of the period and deteriorate over time, with an improvement again in the final year. The reduced performance likely reflects the high level of instability in South America during 2017 and 2018. This period was marked by significant economic volatility, such as Venezuela’s hyperinflation and economic collapse, as well as Argentina’s recession and sharp slowdown in growth. This introduced structural breaks and irregular dynamics that had a big influence on regional migration and were difficult for the model to capture, thereby reducing predictive accuracy.

Focusing on Brazil, Chile, and Argentina allows for a concise yet representative assessment of model performance across different structural contexts.

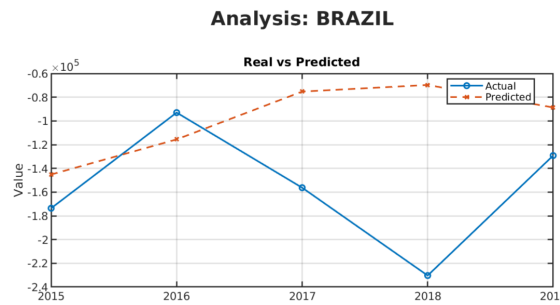


Figure 1: Time Series Brazil

For Brazil, although year-by-year values are not perfectly matched, the predicted series follows the same directional movements observed in the data, even if delayed, correctly identifying phases of increasing and decreasing net outflows. This behavior is consistent with the relatively high explanatory power of the model ($R^2 = 0.76$, Adjusted $R^2 = 0.74$), which indicates that most variation is captured at the structural level, while short-run deviations remain partially unexplained. In the Brazilian context, where migration dynamics evolve gradually and are

closely linked to macroeconomic and labor-market conditions, the error structure implies that the model captures the underlying mechanisms effectively, while smoothing short-term fluctuations. Overall, Brazil represents a case in which the model performs close to its intended scope, functioning as a reliable tool for trend identification and medium-term structural analysis rather than precise annual forecasting [8, 9].

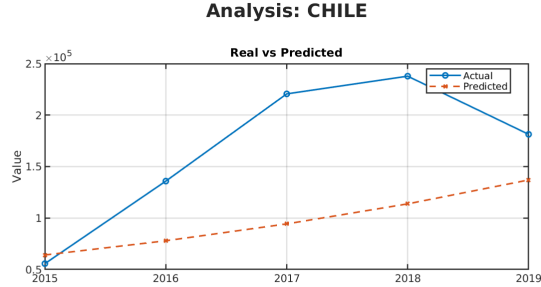


Figure 2: Time Series Chile

A similarly strong performance is observed for Chile. The model accurately captures the persistent positive migration trend over the 2015–2019 period, correctly identifying Chile as a structurally attractive destination within South America. Predicted inflows systematically underestimate the sharp increase observed in the data, particularly during peak years, but the overall trajectory remains highly consistent with observed dynamics. This pattern indicates that the model prioritizes structural determinants of attractiveness, such as relative economic stability and labor-market conditions, while smoothing rapid accelerations linked to regional shocks. The very high accuracy of fit ($R^2 = 0.95$, Adjusted $R^2 = 0.95$) confirms that, in the Chilean case, migration dynamics are strongly aligned with the macroeconomic and institutional variables included in the model. The residual errors therefore reflect a deliberate smoothing of short-term accelerations rather than structural misspecification or misrepresentation of the underlying migration process, reinforcing the model’s effectiveness in capturing structurally driven destination patterns within the region [7].

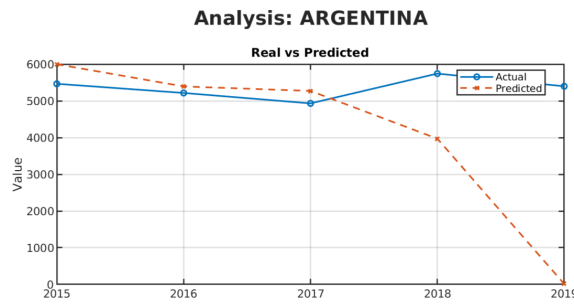


Figure 3: Time Series Argentina

The case of Argentina highlights a key structural sensitivity of the model. Predicted migration aligns closely with observed values in the initial years, but the sharp macroeconomic deterioration starting in 2018 (driven by the GDP contraction and inflation crisis) induces a pronounced downward revision in the predicted series. The extremely high accuracy of fit ($R^2 = 0.995$, Adjusted $R^2 = 0.993$) indicates that migration outcomes in Argentina are overwhelmingly explained by the model’s covariates. As a consequence, the model reacts strongly and rapidly to economic decline, amplifying its impact on expected migration flows and generating a sharper

trend reversal than observed in the data. While this behavior is economically interpretable and consistent with push-factor theories of migration, it suggests that the model may overstate short-term responsiveness to macroeconomic shocks, treating them as immediately binding constraints rather than as signals whose effects materialize with delay [15].

Overall, the results demonstrate that the model performs best as a tool for identifying structural trends and medium-term migration dynamics and lacks precise year-by-year prediction accuracy.

5 Conclusion and Outlook

The refinement of the model marked a fundamental transition from a predominantly autoregressive approach to one based on structural determinants. In its initial version, the system exhibited very high statistical accuracy due to its heavy reliance on temporal inertia. The analysis indicated that the most accurate short-term predictions were derived primarily from the previous year’s net migration and its immediate rate of change (slope/velocity). This suggests that “Migration Inertia” is the strongest indicator of future migration in the short run [11]. However, while relying solely on past values and slopes minimizes error for immediate years, it is structurally unstable for long-term forecasting; such a trend-based model eventually diverges toward positive or negative infinity. Therefore, while the final model incorporates these inertia terms to ensure accuracy, it balances them with structural economic variables to prevent unrealistic divergence over longer horizons.

The model is a tool that, while maintaining solid predictive power, allows for the isolation of the specific contribution of each push-pull variable [7].

At the same time, the Argentine case illustrates an inherent trade-off in structurally driven models: strong responsiveness to macroeconomic indicators, such as GDP, can translate sudden economic downturns into sharp predicted migration shifts that exceed those observed in practice.

The interpretation of the model’s results indicates a capacity for trend identification, capturing the broader directional flows of migration across South America. However, the five-year testing period (2015–2019) is insufficient to draw definitive conclusions about the model’s long-term stability; a longer validation window would be required to confirm these trends. Conversely, the attempt to build a precise year-by-year forecasting model yielded limited success. The high volatility of annual migration flows proved difficult to predict with exact precision, highlighting the inherent unpredictability of the subject matter.

Despite these limitations, the project demonstrated that it is possible to construct a viable predictive model using only five base variables. Given that migration is an extremely complex, multi-causal phenomenon, already achieving a slightly satisfactory performance with such a parsimonious input set is a significant result.

Looking ahead, it would be valuable to apply this modeling framework to more structurally stable regions, such as Europe, to compare performance against the volatile South American context. While further model improvements are certainly possible, particularly with more time to deepen the theoretical understanding of migration dynamics, this work represents a robust implementation within the constraints of the project, establishing a functional baseline for data-driven migration forecasting.

6 Authorship statement

This work is the result of a collaborative effort by the three authors. All authors contributed jointly to the development of the conceptual framework, the research on migration dynamics, the definition of the modeling strategy, and the writing of the report.

Specific contributions are as follows. Mirto Regazzoni developed the MATLAB forecasting model and implemented the full project automation pipeline. Mark Rüegg designed and implemented the C code for results validation, developed the MATLAB visualization routines, and provided support in the implementation of the forecasting model. Aimé Couty was responsible for data research and for the development of the MATLAB preprocessing codes.

Generative AI tools (including ChatGPT, Gemini, and Claude) were used as auxiliary support for debugging, code clarification, and language refinement; all methodological choices, implementations, and interpretations remain the responsibility of the authors.

All authors reviewed and approved the final version of the manuscript and take responsibility for its content.

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A

Definition of Model Variables

This appendix briefly describes the explanatory variables used in the migration model, clarifying their construction and intended interpretation.

GDPgrowth. Annual percentage change in GDP. It captures short-term economic expansion or contraction, reflecting changes in economic opportunities that can influence migration decisions.

GDPaccel. Year-to-year change in GDP growth (second derivative). This variable detects economic shocks and turning points, distinguishing stable growth from sudden slowdowns or recoveries.

LogGDP. Natural logarithm of GDP. The transformation reduces scale effects and captures diminishing returns to economic size, allowing comparisons across countries with very different GDP levels.

LogSchool. Natural logarithm of average years of schooling. This reflects the idea that additional education has decreasing marginal effects on migration incentives at higher levels.

Unemployment Unemployment rate of the labor force. High unemployment acts as a push factor, signaling weak labor-market conditions.

UnempChange. Annual change in the unemployment rate. It captures rapid deteriorations or improvements in labor-market conditions that may trigger migration responses.

Homicide. Homicide rate per 100,000 inhabitants. It serves as a proxy for public safety and social instability, with higher values increasing emigration pressure.

HomicideChange. Yearly change in the homicide rate. This variable reflects sudden increases or decreases in violence rather than long-term crime levels.

LogHomicide. Natural logarithm of the homicide rate. The transformation limits the influence of extreme values and models non-linear effects of violence on migration.

OpportunityIndex. Ratio of GDP to years of schooling. It approximates economic returns to education, capturing how effectively human capital is rewarded within a country.

EconomicStress. Multiplication between unemployment and absolute GDP growth. It represents combined economic pressure, where weak growth and high unemployment reinforce each other as push factors.

SafetyIndex. Multiplication between homicide rate and unemployment. This index captures situations where insecurity and lack of employment jointly intensify migration incentives.

MigrationLag1. Net migration in the previous year. It represents migration inertia and network effects, where past migration facilitates future migration.

MigrationLag2. Net migration two years prior. This longer lag captures more persistent migration dynamics beyond immediate short-term effects.

TimeTrend. Normalized time index ranging from 0 to 1 over the sample period. It accounts for slow-moving structural changes not explicitly captured by other variables.

These variables jointly combine economic dynamics, labor-market conditions, security factors, and migration momentum, providing a structured and interpretable representation of migration behavior.